Ground shakes: a repeatable electrical response

ProtoDUNE-HD · Manuel Arroyave · Analysis through 27 October 2026

We checked seven runs.

2024 data	Run	Events
July	27938	6
	27951	19
October	30201	44
	30202	83
November	32730	1
	32731	716
	32739	662
Total		1,531

Selected records; automatic labels need review. The initiating source is still unknown.

We used charge and PDS data.

TPC: charge readout. PDS: photon detector readout.

Slow controls (SC): 34 monitors; 7.74 million samples.

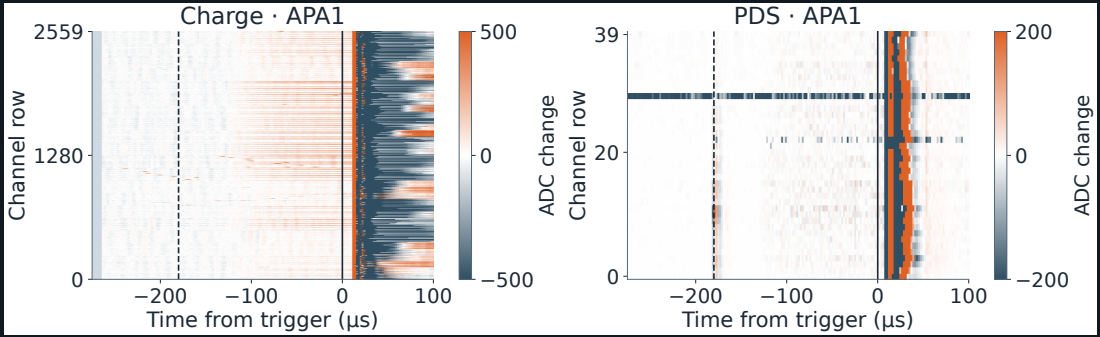
Charge test: 1,516 possible shakes.

Five records did not pass the test; ten lacked usable charge data.

- Same slow charge shape, opposite signs.
- Electrical pattern transfers between runs.
- No robust SC correlation found in our test.

A PDS pulse comes before the large disturbance

Run 27938 · event 20576 · APA1 charge and continuous PDS readout

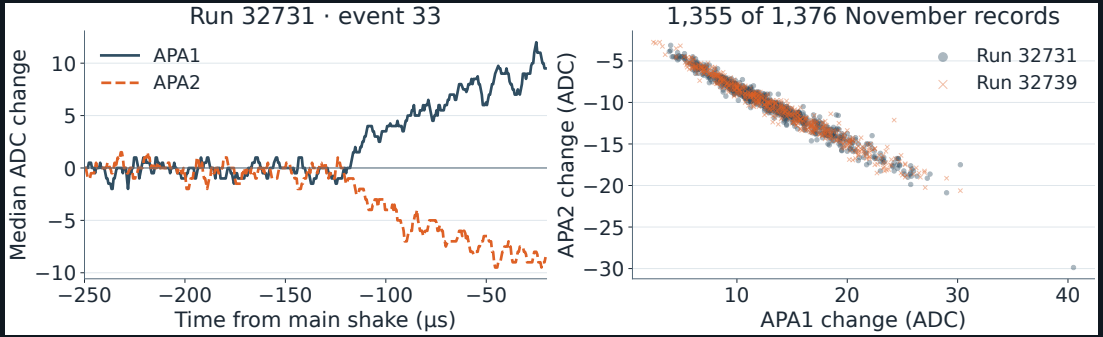


Each row is one channel. Blue is below baseline; orange is above it. The dashed line marks an early PDS pulse, about 180 μs before the trigger (0). Both systems then show a large disturbance. Grey means unrecorded.

ADC counts are raw signal units; μs means millionths of a second. Colour limits clip large signals. Recorded timing; electronics delays uncorrected.

The slow charge signals move in opposite directions

APA1 and APA2 are the two charge-readout assemblies used here

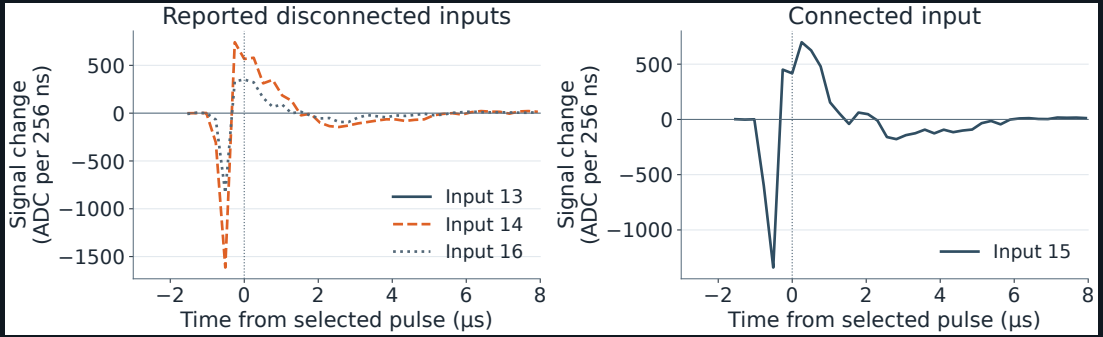


Left: one event. APA1 rises while APA2 falls before the main shake. **Right:** one dot per event. This pattern is clear in 1,355 of 1,376 November records.

Median shape correlation after reversing APA2: 0.964 and 0.970 in the two runs. A score of 1 means the same shape.

Disconnected PDS inputs also carry pulses

Run 32731 · event 33 · selected pulse 183.296 μs before the main shake

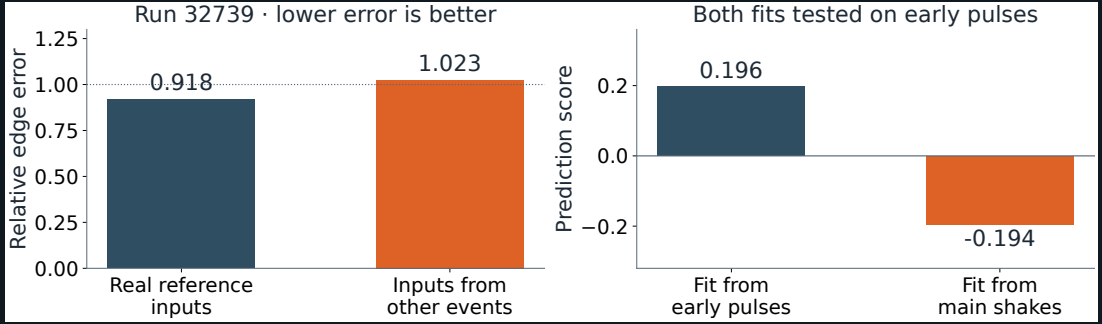


Left: inputs 13, 14 and 16 were reported disconnected from the sensors. **Right:** connected input 15 responds at the same time. This supports electrical pickup. It does not identify what started the pulse.

Each line shows signal change in 256 ns steps. The reported disconnections still need run-specific confirmation.

The electrical pattern transfers to another run

Fit trained on run 32731; all run 32739 events excluded from training

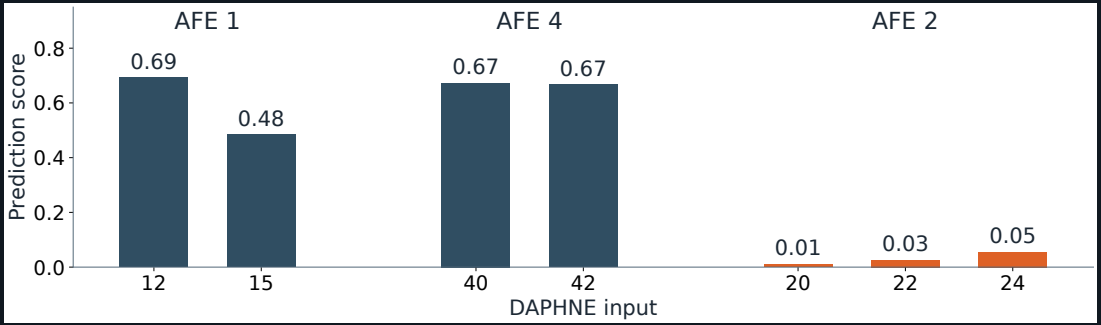


Left: the real disconnected-input signals give less error than inputs from other events. **Right:** a fit trained on the main shake fails on early pulses. One simple response does not describe both stages.

A positive score beats a zero-change guess. Large pulses dominate the combined score; the median score per event is only 0.022. This is not a light/electrical fraction.

The response depends on the electronics channel

Same early-pulse fit; tested on run 32739

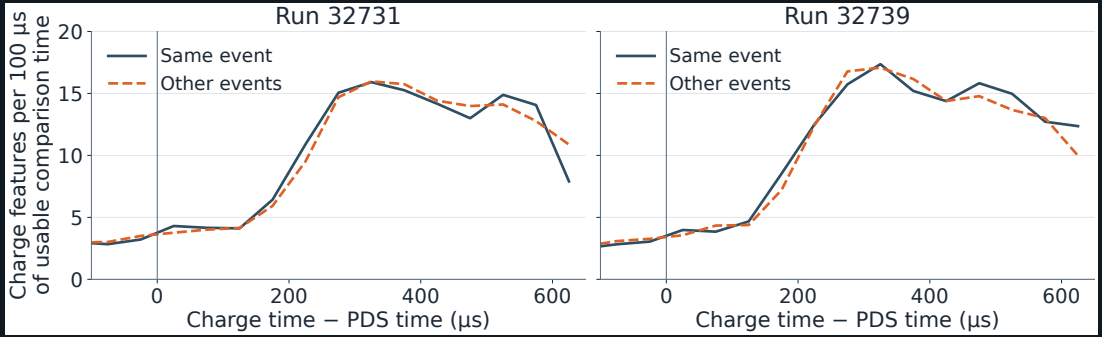


Each bar is one connected input. Taller bars mean better prediction from the disconnected inputs. Inputs 12/15 and 40/42 are well predicted. Inputs 20/22/24 remain largely unexplained.

An AFE is an eight-channel front-end chip. Score: 0 = no improvement; 1 = exact prediction. The unequal response constrains the circuit model.

We found no unique charge–light match

Early PDS pulses compared with later charge features · November runs

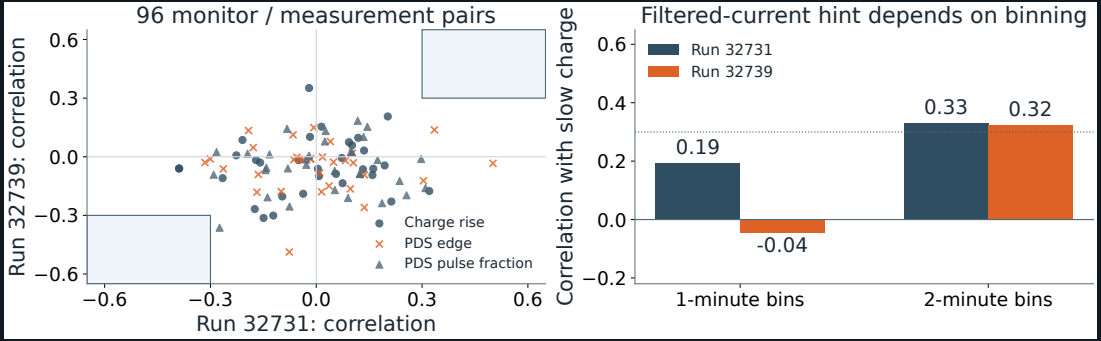


The horizontal axis is charge time minus PDS time. Blue uses pulses from the same event; dashed orange uses pulses from other events. Both show the peak near 325 μs. That peak alone does not link a PDS pulse to a charge track.

Counts account for recorded comparison time. Pulses are swapped within the same run, file and assembly. This is not a normal-running background test.

We found no robust SC–waveform correlation

32 testable monitors × 3 measurements · Runs 32731 and 32739 · Two ground monitors too sparse



Left: each dot tests one monitor against one waveform measurement. Shaded corners mark a repeat in both runs; none qualify with one-minute bins. **Right:** a weak filtered-current hint depends on binning and does not survive the other checks.

Test: magnitude at least 0.30, with the same sign in both runs; file and time-trend adjusted. This tests waveform changes within selected shakes, not shake rate or fast transients.

Next: tracks, controls and the electrical path

USE THE SAVED DATA

Fit charge tracks for 348 unexplained pulse windows

Check the run geometry and timing. Compare whole charge tracks with each possible light pulse.

ADD MATCHED CONTROLS

Measure the excess and false-alarm rate

Find records without shakes using the same boards, gains and triggers, with measured live time.

CHECK THE HARDWARE

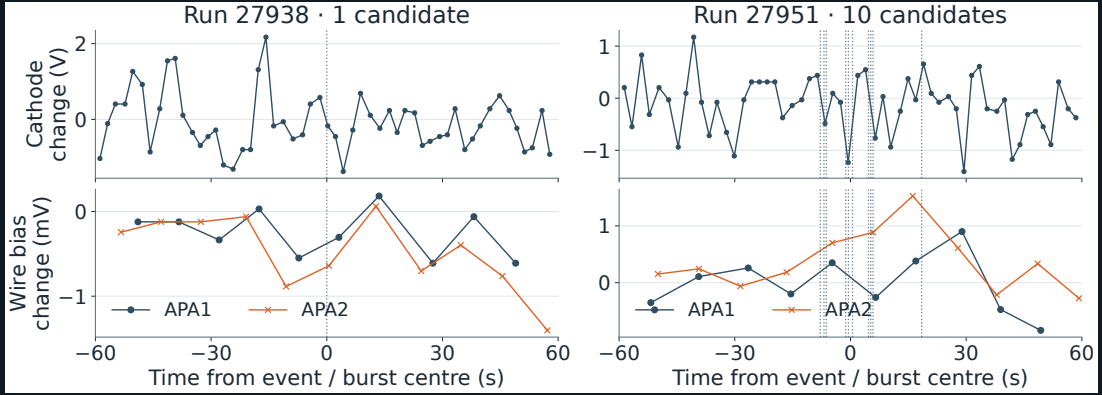
Test which electrical path gives this channel pattern

Confirm wiring and AFE settings. Measure fast supply, ground and reference signals.

No candidate has a full 4 ms of pre-shake PDS data. The SC export contains no direct DAPHNE supply monitors.

Backup: July voltage records

Eleven July candidates; ten occur in one 26-second burst

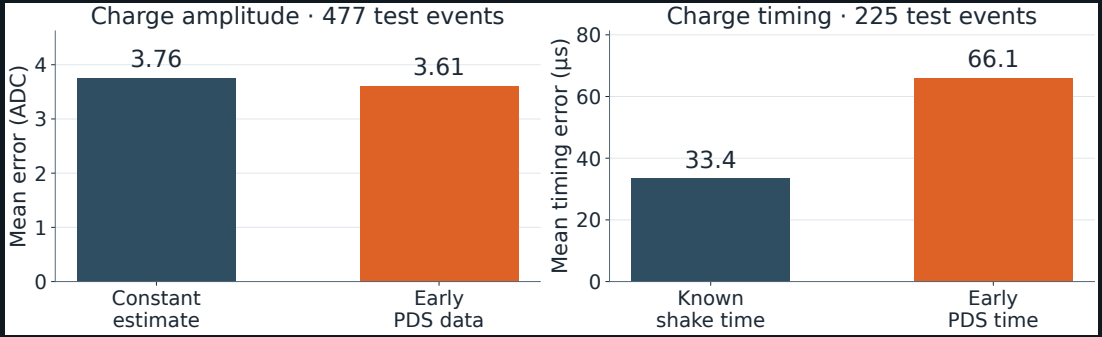


Top: cathode voltage change. **Bottom:** wire-bias change. Dots are archived samples; vertical marks are shake candidates. Sampling every 2–11 seconds cannot rule out a brief discharge.

Changes are relative to the preceding baseline: about 176 kV for the cathode and -370 V for the wire bias.

Backup: early PDS gives weak charge predictions

Trained on run 32731; tested on unused run 32739 events · Lower bars are better



Left: early PDS data reduce the charge-amplitude error by only 4.1%. **Right:** the early pulse predicts the 2 ADC charge crossing less accurately than the known main-shake time. These checks use saved events.

This tests variation within selected shakes. It does not test whether PDS can predict that a shake will occur.